From the Greene–Wu Convolution to Gradient Estimation over Riemannian Manifolds

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Abstract

Over a complete Riemannian manifold of finite dimension, Greene and Wu introduced a convolution of f defined as

$$\widehat{f}^{\mu}(x) := \int_{v \in T_p\mathcal{M}} f\left(\mathrm{Exp}_p(v)\right) \kappa_\mu\left(v\right) d\Pi, \tag{GW}$$

where κ_μ is a kernel that integrates to 1, and $d\Pi$ is the base measure on $T_p\mathcal{M}$. In this paper, we introduce a reformulation of the Greene–Wu (GW) convolution. Using our reformulation, many properties of the GW convolution can be easily derived, including a new formula for how the curvature of the space would affect the curvature of the function through the GW convolution. Also enabled by our new reformulation, an improved method for gradient estimation over Riemannian manifolds is introduced. Theoretically, our gradient estimation method improves estimation error bound from $\frac{L_1(n+3)^{3/2}\mu}{2}$ to $\frac{L_1 n^{3/2}\mu}{2}$, where n is the dimension of the manifold, L_1 is the smoothness parameter of the objective function and μ is an algorithm parameter controlling finite-difference step size. Empirically, our method outperforms the best existing method for gradient estimation over Riemannian manifolds, as evidenced by thorough experimental evaluations.

1 Introduction

In their seminal papers, Greene and Wu studied the GW convolution

$$\widehat{f}^{\mu}(p) := \int_{v \in T_p\mathcal{M}} f\left(\mathrm{Exp}_p(v)\right) \kappa_\mu\left(v\right) d\Pi, \tag{1}$$

where κ_μ is a radial convolution kernel that integrates to 1, and $d\Pi$ is the base measure on $T_p\mathcal{M}$.

In this paper, we introduce a new formulation of the GW convolution (also known as GW approximation) for radial kernels, and study properties and applications of the GW convolution based on our formulation. To define the value of the GW convolution of f at p over $\mathcal{M}$, one needs the functions values of f over the entire image of the tangent space $T_p\mathcal{M}$ and a reweighting kernel κ_μ. To naturally use the functions values over the entire image of $T_p\mathcal{M}$ for a single point p, one can focus on $T_p\mathcal{M}$ and carry out computations in this tangent space, as is done in the original definition (GW). However, this definition makes it difficult to factor in the geometric structure of the manifold. The reason behind this difficulty is that, once the attention is restricted to the tangent space, intensive computations (e.g., expansions involving Christoffel symbols) become necessary if one wants to depict the geometric structure. Such computations may become wildly intractable, especially when a convolution is involved. To this end, we give a reformulation of the GW convolution, so that the geometric structure is built into the convolution itself, and computations and applications involving the GW convolution can be easily implemented. Naturally, the radial convolution can be decomposed into two steps. In the first step a direction v is chosen and a convolution is performed along this direction. In the second step, the direction v is randomized over a sphere so that the convolution over the entire image of $T_p\mathcal{M}$ is recovered.

Following this intuition, we extend the first step to factor in the curvature of the space. Specifically, in the first step where a direction v is picked, the convolution need not be only locally defined. Instead, given a point p and $v \in T_p\mathcal{M}$, we can define the first-step convolution to be $\widetilde{f}^{\mu}_v(q) = \int_{\mathbb{R}} 2f\left(\mathrm{Exp}_q(tv_q)\right)\Theta_\mu(t)dt, \quad \forall q \in \mathcal{M}$, where Θ_μ is the density along $\mathbb{R}$ induced by κ_μ, and v_q is the vector from parallel transporting v from p to q

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along the minimizing geodesic*. This decomposition operation allows the first-step convolution to be computed at points other than p, without resorting to any heavy computational machineries. Surprisingly, this simple decomposition trick can significantly simplify computations and leads to meaningful applications. To illustrate the power of our new formulation, we (1) prove the convexity-preserving property of the GW convolution (Greene and Wu, 1973, 1976, 1979) with a succinct argument, and (2) derive a new formula for how the curvature of the space $\mathcal{M}$ would affect the curvature of the function f via the GW convolution.

Also enabled by our new decomposition of the GW convolution, we improve previous studies on the gradient estimation over Riemannian manifolds. For such tasks, we improve the best known result (Li et al., 2020) by establishing a tighter error bound: For estimating the gradient of smooth functions over an n-dimensional Riemannian manifold, we show the estimation error is bounded by $\frac{L_1(n+3)^{3/2}\mu}{2}$ instead of $\frac{L_1 n^{3/2}\mu}{2}$, where n is the dimension of the manifold, L_1 is the smoothness parameter of the objective function and μ is an algorithm parameter controlling finite-difference step size. The supremacy of our method is evidenced by a thorough empirical study. Another advantage of our results is that they do not rely on an ambient Euclidean space, which is more general than the previous work (Li et al., 2020).

We also apply our estimation method to online convex learning over Hadamard manifolds. In particular, single point gradient estimator is designed, and thus online convex learning problem can be solved under bandit feedback.

Related Works

Our work is related to function approximation and convolution over Riemannian manifolds. Pioneering works for such problems include the seminal papers by Greene and Wu (1973, 1976, 1979). In their works, a convolution of the form $\widehat{f}^{\mu}(p) := \int_{v\in T_p\mathcal{M}} f\left(\mathrm{Exp}_p(v)\right)\kappa_\mu(v)\,d\Pi$ is studied, where $\int_{v\in\mathbb{R}^n}\kappa_\mu(v)\,dv = 1$ is the convolution kernel and $d\Pi$ is the base measure on $T_p\mathcal{M}$ (induced by the metric on $\mathcal{M}$ via the exponential map). Later, Parkkonen and Paulin (2012) used this convolution scheme for handling regularity conditions in studying convex sets in negatively curved manifolds. Also, Azagra and Ferrera (2006); Azagra (2013) generalized the GW convolution using modern languages. In particular, they study smooth approximation of functions in the C^k-fine topology, and show that over real manifolds, the approximation properties of f is determined by its global structure, rather than its local behavior. Previous works on the GW convolution have evidenced its wide applicability, which urgently calls for better computational tools for studying the GW convolution. In this paper, we provide such tools by directly reformulate the GW convolution, so that the computations and applications of the GW convolution can be easily implemented.

Our work implies advancements in gradient estimation, whose history can date back to Sir Isaac Newton and the fundamental theory of calculus. Since then, many gradient estimation methods have been invented (Lyness and Moler, 1967; Hildebrand, 1987). In its more recent form, gradient estimation methods are often combined with learning and optimization. Flaxman et al. (2005) used the fundamental theorem of geometric calculus and solved online convex optimization problems with bandit feedback. Nesterov and Spokoiny (2017); Li et al. (2020) introduced a stochastic gradient estimation method using Gaussian sampling, in Euclidean space or manifolds embedded in Euclidean spaces. We improve the previous results in two ways. 1. Our gradient estimation method generalizes this previous work by removing the ambient Euclidean space. 2 We tighten the previous error bound. Over an n-dimensional Riemannian manifold, we show that the error of gradient estimation is bounded by $\frac{L_1(n+3)^{3/2}\mu}{2}$ in contrast to $\frac{L_1 n^{3/2}\mu}{2}$(Li et al., 2020), where n is the dimension of the manifold, L_1 is the smoothness parameter of the objective function and μ is an algorithm parameter controlling finite-difference step size.

Notations and Conventions

Before proceeding, we list some notations and conventions that will be used throughout the rest of this paper for better readability.

- For any $x \in \mathcal{M}$, let U_p denote the open set near p that is a diffeomorphic to a subset of $\mathbb{R}^n$ via the local normal coordinate chart ϕ. Define the distance $d_p(q_1, q_2)$ $(q_1, q_2 \in U_p)$ such that

$$d_p(q_1, q_2) = d_{\mathrm{Euc}}(\phi(q_1), \phi(q_2)). \tag{2}$$

  where d_{Euc} is the Euclidean distance in $\mathbb{R}^n$.

*When the minimizing geodesic is non-unique, one can simply uniformly randomly pick one minimizing geodesic.

- For any $p \in \mathcal{M}$, we use $\langle \cdot, \cdot \rangle_p$ and $\| \cdot \|_p$ to denote the inner produce and norm induced by the Riemannian metric g in the tangent space $T_p\mathcal{M}$. We omit the subscript when it is clear from context.
- For any $p \in \mathcal{M}$ and differentiable function f over $\mathcal{M}$, We use $\mathrm{grad} f\big|_p$ to denote the gradient of f at p.
- For any $p \in \mathcal{M}$ and $\alpha > 0$, we use $\mathbb{S}_p(\alpha)$ to denote the origin-centered sphere in $T_p\mathcal{M}$ with radius α. For simplicity, we write $\mathbb{S}_p = \mathbb{S}_p(1)$.
- For two points $p, q \in \mathcal{M}$ and a smooth curve γ connecting p and q, we use $\mathcal{P}^{\gamma}_{p\to q}$ to denote the parallel transport from $T_p\mathcal{M}$ to $T_q\mathcal{M}$ along the curve γ. Let $\Omega_{\min}(p, q)$ denote the set of minimizing geodesics connecting p and q. Define $\mathcal{P}_{p\to q} : T_p\mathcal{M} \to T_q\mathcal{M}$ be defined such that $\mathcal{P}_{p\to q}(v) = \mathbb{E}_{\gamma\sim\Omega_{\min}(p,q)} \left[\mathcal{P}^{\gamma}_{p\to q}(v)\right]$ for any $v \in T_p\mathcal{M}$, where the expectation is taken with respect to the uniform probability measure over $\Omega_{\min}(p, q)$. Since for any point p, its cut locus is of measure zero, the operation $\mathcal{P}_{p\to q}$ is deterministically defined for almost every q.
- For any $p \in \mathcal{M}$, $v \in T_p\mathcal{M}$ and $q \in \mathcal{M}$, define $v_q = \mathcal{P}_{p\to q}(v)$.
- We will use another set of notations for parallel transport along curves. For a smooth curve $\gamma : [a, b] \to \mathcal{M}$, we use $\mathcal{P}^{\gamma}_{c,d}$ to denote the parallel transport from $T_{\gamma(c)}\mathcal{M}$ to $T_{\gamma(d)}\mathcal{M}$ along γ.
- As a convention, if $U \subseteq \mathcal{M}$ is diffeomorphic to $V \subseteq \mathbb{R}^n$, we simply say U is diffeomorphic. Similarly, a map is said to be diffeomorphic when the domain of the map is diffeomorphic to an open subset of the Euclidean space. For simplicity, we omit all musical isomorphisms when there is no confusion.

2 The Greene–Wu Convolution

The Greene–Wu convolution with radial kernels, can be decomposed into two steps. In the first step, a direction v in the tangent space is picked, a convolution along the direction is performed, and the extension of this convolution to other points is defined. In the second step, the direction v is randomized over a sphere. To properly describe these two steps, we need to formally define radial kernels. A kernel $\kappa_\mu : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ is called a radial kernel if

1. For any $\alpha > 0$ and $\mu > 0$, there exists a density $\Theta_{\mu,\alpha}$ on $\mathbb{R}$ such that $\kappa_\mu(v) = 2\nu_\alpha \left(\alpha \frac{v}{\|v\|}\right) \Theta_{\mu,\alpha} \left(\frac{\|v\|}{\alpha}\right)$, and $\Theta_{\mu,\alpha}(-x) = \Theta_{\mu,\alpha}(x)$ for all $x \in \mathbb{R}$, where ν_α is the uniform probability density over the Euclidean sphere of radius α[†];
2. $\lim_{\mu\to 0^+} \kappa_\mu = \delta_0$, where δ_0 is the Dirac point mass at 0.

By the Radon-Nikodym theorem, item 1 in the definition of a radial kernel is satisfied when the measure on $T_p\mathcal{M}$ induced by κ_μ is absolutely continuous with respect to the base measure on $T_p\mathcal{M}$. Examples of radial kernels include uniform mass over a ball of radius μ.

We now formally define the GW approximation/convolution. For $p \in \mathcal{M}$ and $\mu > 0$, we define

$$\widetilde{f}^{\mu}(p) := \mathbb{E}_{v\sim\mathbb{S}_p(\alpha)} \left[\widetilde{f}^{\mu}_v(p)\right], \tag{3}$$

where

$$\widetilde{f}^{\mu}_v(q) = \int_{\mathbb{R}} 2f\left(\mathrm{Exp}_q(tv_q)\right) \Theta_{\mu,\alpha}(t)dt, \quad \forall q \in \mathcal{M}. \tag{4}$$

The difference between (GW) and (3) is more subtle than replacing the integration in (GW) with a radial integration. In particular, the function in (4) is defined for all $q \in \mathcal{M}$, instead of only at p. This extension of the convolution along v can significantly simplify computations and implies critical applications. Next we show that the convolution in (3) is identical to the (GW) convolution everywhere.

Proposition 1. *For any $p \in \mathcal{M}$ and κ_μ defined on $T_p\mathcal{M} \cong \mathbb{R}^n$, it holds that $\widetilde{f}^{\mu}(p) = \widehat{f}^{\mu}(p)$.*

[†]Here $\kappa_\mu(0) = 2\Theta_{\mu,\alpha}(0)$ as a convention.

Proof. Let dt be the base measure on $\mathbb{R}$ and let dS_α be the base measure on $\mathbb{S}_p(\alpha)$. We then have $dtdS_\alpha = d\Pi$. Thus it holds that

$$\widetilde{f}^\mu(p) = \int_{v\in\mathbb{S}_p(\alpha)} \int_{t\in\mathbb{R}} 2f\left(\mathrm{Exp}_p(tv)\right) \nu_\alpha\left(\alpha\frac{v}{\|v\|}\right) \Theta_{\mu,\alpha}(t)\, dtdS_\alpha = \int_{u\in T_p\mathcal{M}} f\left(\mathrm{Exp}_p(u)\right) \kappa_\mu(u)\, d\Pi = \widehat{f}^\mu(p),$$

where a change of variable $u = tv$ is used.

□

Remark 1. *With the Levi-Civita connection, spheres are preserved after parallel transport, and so are the uniform base measures over the spheres. Thus the spherical integration of $\widetilde{f}^\mu_v$ defined in (4) can be carried out at p regardless whether $v \in T_p\mathcal{M}$ or not. Hence, the convolution defined by (3) and (4) is indeed the GW convolution in the sense that $\widetilde{f}^\mu_v$ agrees with (GW) almost everywhere*[‡] *if the convolution at every point is restricted to the direction specified by v (or parallel transport of v). Now the advantage of our decomposition is clear – $\widetilde{f}^\mu_v$ is globally defined and agrees with (GW) along the the direction of v. This advantage leads to easy computations and meaningful applications, as we will see throughout the rest of the paper.*

2.1 Properties of the GW Convolution

Previously, the convexity-preserving properties had been a focus of the studies by Greene and Wu. Specifically, Greene and Wu showed that the approximation is convexity-preserving asymptotically (Theorem 1). Using our new reformulation of the GW approximation, we provide an alternative proof to the convexity-preserving results of Greene and Wu. Also enabled by our reformulation, second order properties of the GW approximation is derived.

Definition 1 (Convexity)**.** *Let $\mathcal{M}$ be a complete Riemannian manifold. A smooth function f is convex near p with radius ϵ if there exists $\epsilon > 0$, such that (1) for any $v \in \mathbb{S}_p$, the function $f\left(\mathrm{Exp}_p(tv)\right)$ is convex in t over $[-\epsilon, \epsilon]$, (2) there exists $v \in \mathbb{S}_p$ such that $f\left(\mathrm{Exp}_p(tv)\right)$ is not convex in t over $[-\epsilon-\epsilon', \epsilon+\epsilon']$ for any $\epsilon' > 0$.*

Theorem 1 (Greene-Wu)**.** *Consider a smooth function defined over a complete Riemannian manifold $\mathcal{M}$. Let $\widehat{f}^\mu$ be the GW approximation of f with kernel κ_μ. Then it holds that*

$$(i)\ \lim_{\mu\to 0^+} \widehat{f}^\mu(p) = f(p) \quad \textit{and} \quad (ii)\ \liminf_{\mu\to 0^+}\left(\inf_{C\in\mathcal{C}_p} \frac{d^2\widehat{f}^\mu(C(t))}{dt^2}\Big|_{t=0}\right) \geq 0, \tag{5}$$

where $\mathcal{C}_p$ is the set of all geodesics C parametrized by arc-length such that $C(0) = p$.

Theorem 1 shows that $\widehat{f}^\mu$ is a good approximation of f by item (i), and shows that $\widehat{f}^\mu$ is asymptotically convexity-preserving by item (ii). In proving item (ii), Greene and Wu showed that, for any curve C, $\widehat{f}^\mu(C(\lambda)) + \widehat{f}^\mu(C(-\lambda)) \geq 2\widehat{f}^\mu(C(0)) - \xi\lambda^2$ for some $\mu \geq 0$ and this ξ vanishes as μ goes to zero.

Using our decomposition trick of the GW approximation (defined in Eq. 3), one can give an alternative proof for the convexity-preserving property of the GW convolution.

Theorem 2. *Let $\mathcal{M}$ be a complete Riemannian manifold, and fix an arbitrary $p \in \mathcal{M}$. If there exists $\epsilon > 0$ such that 1) f is convex near p with radius ϵ, and 2) the density Θ_μ (a shorthand for $\Theta_{\mu,1}$) induced by kernel κ_μ satisfies that $\int_{|t|\geq\frac{\epsilon}{2}} \Theta_\mu(t) = 0$, then the Greene-Wu approximation $\widehat{f}^\mu$ is also convex near p.*

Proof of Theorem 2. By Proposition 1 and Remark 1, it is sufficient to consider $\widetilde{f}^\mu$. Since f is convex near p with radius ϵ, we have, for any $v \in T_p\mathcal{M}$, there exists $\epsilon > 0$ such that for all real numbers $\lambda, t \in \left(0, \frac{\epsilon}{2}\right)$, it holds that

$$f(\mathrm{Exp}_p((\lambda+t)v)) + f(\mathrm{Exp}_p((-\lambda+t)v)) \geq 2f(\mathrm{Exp}_p(tv)). \tag{6}$$

Let $\gamma(s) = \mathrm{Exp}_p(sv)$ ($s \in [-\lambda-t, \lambda+t]$) for simplicity. We can then rewrite (6) as

$$f(\mathrm{Exp}_{\gamma(\lambda)}(t\mathcal{P}^\gamma_{0,\lambda}(v))) + f(\mathrm{Exp}_{\gamma(-\lambda)}(t\mathcal{P}^\gamma_{0,-\lambda}(v))) \geq 2f(\mathrm{Exp}_p(tv)) \tag{7}$$

We can integrate the first term in (7) with respect to v and t, and get

$$\int_{v\in\mathbb{S}_p}\int_{t\in\left[0,\frac{\epsilon}{2}\right]} 2f(\mathrm{Exp}_{\gamma(\lambda)}(t\mathcal{P}^\gamma_{0,\lambda}(v)))\nu_1(v)\Theta_\mu(t)dtdS_1 = \int_{v\in\mathbb{S}_{\gamma(\lambda)}}\int_{t\in\left[0,\frac{\epsilon}{2}\right]} 2f(\mathrm{Exp}_{\gamma(\lambda)}(tv))\nu_1(v)\Theta_\mu(t)dt\, dS_1, \tag{8}$$

[‡]The cut locus is of measure zero and thus v_q is deterministically defined almost everywhere.

where (8) uses that parallel transport along geodesics preserves length and angle, for the change of tangent space for the outer integral.

Since the density Θ_μ satisfies that $\int_{|t|\geq \frac{\epsilon}{2}} \Theta_\mu(t) = 0$, the term on the right-hand-side of (8) is the exact definition of $\widetilde{f}^\mu(\gamma(\lambda))$. We repeat this computation for the second term in (7), and get

$$\widetilde{f}^\mu(\gamma(\lambda)) + \widetilde{f}^\mu(\gamma(-\lambda)) \geq 2\widetilde{f}^\mu(\gamma(0)), \quad \forall \lambda \in \left[0, \frac{\epsilon}{2}\right], \tag{9}$$

which is the midpoint convexity (around $\gamma(0)$, along γ) for $\widetilde{f}^\mu$. Since midpoint convexity implies convexity for smooth functions, we know $\widetilde{f}^\mu$ is convex near p along γ.

□

Our decomposition of the Greene-Wu convolution also allows other properties of the GW convolution easily derived. Perhaps one of the most important property of a function is its Hessian, which we discuss now. The definition of Hessian is in Definition 2 (e.g., Schoen and Yau, 1994; Petersen et al., 2006).

Definition 2 (Hessian)**.** *Consider a Riemannian manifold $(\mathcal{M}, g)$. For $f \in C^\infty(\mathcal{M})$, define the Hessian of f as $H(f) : \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \to C^\infty(\mathcal{M})$ such that*

$$H(f)\,(X, Y) := \nabla_X df(Y), \quad \forall X, Y \in \mathfrak{X}(\mathcal{M}),$$

where ∇ is the Levi-Civita connection. In local coordinates, it holds that $H(f)(\partial_i, \partial_j) = \partial_i\partial_j f - \Gamma^k_{ij}\partial_k f$. Also in local coordinates, the Hessian matrix of f at p is $H_{p,f} = \left[\partial_i\partial_j f\big|_p\right]$.

With the Levi-Civita connection, which is torsion-free, one can verify that $H(f)(X, Y) = H(f)(Y, X)$. For any $p \in \mathcal{M}$, with the normal coordinate specified by the exponential map Exp_p, the Christoffel symbol vanishes at p. This means $H(f)(\partial_i, \partial_j)\Big|_p = \partial_i\partial_j f\big|_p$, which is the entries in the Hessian matrix at p. Similar to the Euclidean case, the eigenvalues of the Hessian describes the curvature of the function. Next, we will show how the curvature of the function is affected after applying the GW convolution.

In the study of the second order properties of the GW approximation, we will restrict our attention to kernels whose density function is uniform. More specifically, we will use $\alpha = 1$ and $\Theta_{\mu,\alpha}(t) = \frac{1}{2\mu}\mathbb{I}_{[t\in[-\mu,\mu]]}$. Since one can construct other radial kernels by combining different uniform density functions, investigating kernels with uniform density function is sufficient for our purpose. With this uniform density function, we have

$$\widetilde{f}^\mu_v(q) = \frac{1}{2\mu}\int_{-\mu}^{\mu} f\left(\mathrm{Exp}_q(tv_q)\right) dt. \tag{10}$$

The following theorem tells us how the geometry of the function and the geometry of the space are related after the GW convolution.

Theorem 3. *Consider a complete Riemannian manifold $\mathcal{M}$. For any $p \in \mathcal{M}$, there exists $\mu_0 > 0$ such that for all $\mu \in (0, \mu_0)$, it holds that*

$$D_{uu}\widetilde{f}^\mu_v(p) = \frac{1}{2\mu}\int_{t\in[-\mu,\mu]} D_{u_{\mathrm{Exp}_p(tv)}u_{\mathrm{Exp}_p(tv)}} f(\mathrm{Exp}_p(tv)) + \frac{1}{2}\mu^3\left\langle \mathrm{grad} f\big|_p, \nabla_v R(v,u)u + \nabla_u R(v,u)v\right\rangle + O\left(\mu^5\right),$$

for all $v, u \in \mathbb{S}_p$, where R is the Riemann curvature tensor.

A tool needed for Theorem 3 is the double exponential map (Gavrilov, 2007). The double exponential map (also written Exp_p) is defined as $\mathrm{Exp}_p : T_p\mathcal{M} \times T_p\mathcal{M} \to \mathcal{M}$ such that, when $v \in T_p\mathcal{M}$ is small enough ($\mathrm{Exp}_p(v) \in U_p$),

$$\mathrm{Exp}_p(v, u) = \mathrm{Exp}_{\mathrm{Exp}_p(v)}(u_{\mathrm{Exp}_p(v)}). \tag{11}$$

Since the exponential maps are smooth, we can define a smooth function $h_p : T_p\mathcal{M} \times T_p\mathcal{M} \to T_p\mathcal{M}$, such that $\mathrm{Exp}_p(h_p(v, u)) = \mathrm{Exp}_p(v, u)$. For small and fixed v and u, one can introduce variables $(t_1, t_2) \in \mathbb{R}^2$ and let $h_p(u, v) = h_p(t_1u, t_2v)\big|_{(t_1,t_2)=(1,1)}$. One can then expand the h_p around $(0, 0)$ and get the double Taylor series. More specifically, one can expand $h_p(u, v)$ by $h_p(u, v) = \sum_{k=0}^\infty \sum_{l=0}^\infty \frac{1}{k!l!}\frac{d^{k+l}}{ds_1^k ds_2^l} h_p(s_1u, s_2v)\Big|_{(0,0)}$. Gavrilov (2007) has computed the series expansion of the h_p function up to the fifth order:

$$\begin{aligned} h_p(u, v) =& u + v + \frac{1}{6}R(v, u)u + \frac{1}{3}R(v, u)v + \frac{1}{12}\nabla_u R(v, u)u + \\ & \frac{1}{24}\nabla_v R(v, u)u + \frac{5}{24}\nabla_u R(v, u)v + \frac{1}{12}\nabla_v R(v, u)v + \cdots, \end{aligned} \tag{12}$$

where $\cdots$ omits terms of degree five and higher.

By comparing (12) with $h_p(u,v) = \sum_{m=0}^{\infty}\sum_{k=0}^{\infty} \frac{1}{k!l!}\frac{d^{k+l}}{ds_1^k ds_2^l} h_p(s_1u, s_2v)\Big|_{(0,0)}$, one has

$$\frac{d^4}{ds_1^2 ds_2^2} h_p(s_1u, s_2v)\Big|_{(0,0)} = \frac{d^4}{ds_1^2 ds_2^2} h_p(-s_1u, s_2v)\Big|_{(0,0)} = \frac{1}{24}\nabla_v R(v,u)u + \frac{5}{24}\nabla_u R(v,u)v \tag{13}$$

$$\frac{d^4}{ds_1^2 ds_2^2} h_p(s_2v, s_1u)\Big|_{(0,0)} = \frac{d^4}{ds_1^2 ds_2^2} h_p(s_2v, -s_1u)\Big|_{(0,0)} = \frac{1}{24}\nabla_u R(u,v)v + \frac{5}{24}\nabla_v R(u,v)u. \tag{14}$$

With the double exponential map and the equalities in (13) and (14), we can prove Theorem 3.

Proof of Theorem 3. With this notion of double exponential map and the h_p function, we have

$$\begin{aligned}
&D_{uu}\widetilde{f}_v^{\mu}(p)\\
=&\lim_{\tau\to 0}\frac{\widetilde{f}_v^{\mu}(\mathrm{Exp}_p(\tau u)) - 2\widetilde{f}_v^{\mu}(p) + \widetilde{f}_v^{\mu}(\mathrm{Exp}_p(-\tau u))}{\tau^2}\\
=&\frac{1}{2\mu}\int_{t\in[-\mu,\mu]}\lim_{\tau\to 0}\frac{f(\mathrm{Exp}_{\mathrm{Exp}_p(\tau u)}(tv_{\mathrm{Exp}_p(\tau u)})) - 2f(\mathrm{Exp}_p(tv)) + f(\mathrm{Exp}_{\mathrm{Exp}_p(-\tau u)}(tv_{\mathrm{Exp}_p(-\tau u)}))}{\tau^2}\\
=&\frac{1}{2\mu}\int_{t\in[-\mu,\mu]}\lim_{\tau\to 0}\frac{f(\mathrm{Exp}_p(\tau u, tv)) - 2f(\mathrm{Exp}_p(tv)) + f(\mathrm{Exp}_p(-\tau u, tv))}{\tau^2}\\
=&\frac{1}{2\mu}\int_{t\in[-\mu,\mu]}\lim_{\tau\to 0}\frac{f(\mathrm{Exp}_p(h_p(\tau u, tv))) - 2f(\mathrm{Exp}_p(tv)) + f(\mathrm{Exp}_p(h_p(-\tau u, tv)))}{\tau^2}.
\end{aligned} \tag{15}$$

Let $\vartheta = f\circ \mathrm{Exp}_p$, let $\varphi_u(t,\tau) = \vartheta(h_p(tv,\tau u))$, and let

$$\psi_{u,v}(\tau,t) = \vartheta(h_p(\tau u, tv)) - \vartheta(h_p(tv,\tau u)) + \vartheta(h_p(-\tau u, tv)) - \vartheta(h_p(tv, -\tau u)).$$

We can then write (15) as

$$D_{uu}\widetilde{f}_v^{\mu}(p) = \frac{1}{2\mu}\int_{t\in[-\mu,\mu]}\lim_{\tau\to 0}\frac{\varphi_u(t,\tau) - 2\varphi_u(t,0) + \varphi_u(t,-\tau) + \psi_{u,v}(\tau,t)}{\tau^2}. \tag{16}$$

For the terms involving φ_u, one has

$$\lim_{\tau\to 0}\frac{\varphi_u(t,\tau) - 2\varphi_u(t,0) + \varphi_u(t,-\tau)}{\tau^2} = \frac{d^2\varphi_u(t,\tau)}{d\tau^2}\Big|_{\tau=0} = D_{u_q u_q} f(q), \tag{17}$$

where $q = \mathrm{Exp}_p(tv)$ and $u_q = \mathcal{P}_{p\to q}(u)$.

For the term $\psi_{u,v}(\tau,t)$, we can Taylor expand it around $(0,0)$ and get

$$\psi_{u,v}(\tau,t) = \sum_{k=0}^{\infty}\sum_{l=0}^{\infty}\frac{\tau^k t^l}{k!l!}\frac{d^{k+l}\psi_{u,v}(s_1,s_2)}{ds_1^k ds_2^l}\Big|_{s_1=0,s_2=0}.$$

Since $\psi_{u,v}(0,t) = 0$ and $\psi_{u,v}(\tau,t)$ is even in τ, the terms that do not depend on τ or are linear in τ vanish. Thus the $\psi_{u,v}(\tau,t)$ can be written as

$$\psi_{u,v}(\tau,t) = \frac{1}{4}\tau^2 t^2\frac{d^4\psi_{u,v}(s_1,s_2)}{ds_1^2 ds_2^2}\Big|_{s_1=0,s_2=0} + O(\tau^2 t^4) + O(\tau^3) + O(\tau^2)\cdot Odd(t), \tag{18}$$

where $Odd(t)$ denotes terms that are odd in t.

By chain rule, one has

$$\frac{d^4\psi_{u,v}(s_1,s_2)}{ds_1^2 ds_2^2}\Big|_{s_1=0,s_2=0} = \left\langle \mathrm{grad}\vartheta\big|_0, \frac{d^4}{ds_1^2 ds_2^2}\left(h_p(s_1u,s_2v) - h_p(s_2v,s_1u) + h_p(-s_1u,s_2v) - h_p(s_2v,-s_1u)\right)\Big|_{(0,0)}\right\rangle.$$

Combining equations (17), (16), (18), (13), (14), and that $\int_{t\in[-\mu,\mu]} Odd(t) = 0$ gives

$$D_{uu}\widetilde{f}_v^{\mu}(p) = \frac{1}{2\mu}\int_{t\in[-\mu,\mu]} D_{u_q u_q} f(q) + \frac{1}{2}\mu^3\left\langle \mathrm{grad}\vartheta\big|_0, \nabla_v R(v,u)u + \nabla_u R(v,u)v\right\rangle + O\left(\mu^5\right), \tag{19}$$

where $q = \mathrm{Exp}_p(tv)$. If we let Exp_p define a local coordinate at p and let $r(x) = d_p(p, x)$ for all $x \in U_p$, then the metric g satisfies $g_{ij} = \delta_{ij} + O(r^2)$ (See e.g., Petersen et al., 2006). Thus we have $\mathrm{grad} f\big|_p = \mathrm{grad}\vartheta\big|_0$, since the $O(r^2)$ term is second-order small and does not contribute to gradient. Plugging that $\mathrm{grad} f\big|_p = \mathrm{grad}\vartheta\big|_0$ into (19) finishes the proof.

□

Theorem 3 says that when the space is not flat, the GW convolution changes the curvature of the function in a non-trivial way. A more concrete example of this fact is in Corollary 1.

Corollary 1. *For a smooth function f defined over a complete Riemannian manifold $\mathcal{M}$ and any $p \in \mathcal{M}$, there exists $\mu_0 > 0$, such that for all $\mu \in (0, \mu_0)$, it holds that*

$$\begin{aligned}\lambda_{\min}\left(H_{p,\widetilde{f}^\mu}\right) \geq & \frac{1}{2\mu}\mathbb{E}_{v\sim\mathbb{S}_p}\left[\int_{-\mu}^{\mu}\lambda_{\min}\left(H_{\mathrm{Exp}_p(tv),f}\right)dt\right] \\ & + \min_{u\in\mathbb{S}_p}\frac{1}{2}\mu^3\mathbb{E}_{v\sim\mathbb{S}_p}\left[\left\langle \mathrm{grad} f\big|_p, \nabla_v R(v,u)u + \nabla_u R(v,u)v\right\rangle\right] + O\left(\mu^5\right).\end{aligned}$$

Since the minimum eigenvalue of the Hessian matrix measures the degree of convexity of a function, this corollary establish a quantitative relative of how the GW convolution would affect the degree of convexity.

Proof of Corollary 1. For any smooth function f and point p, one has

$$u^\top H_{p,f} u = u^i u^j \partial_i \partial_j f\big|_p = D_{uu} f(p), \tag{20}$$

and thus

$$\lambda_{\min}\left(H_{p,\widetilde{f}^\mu}\right) = \min_{u\in\mathbb{S}_p} u^\top H_{p,\widetilde{f}^\mu} u = \min_{u\in\mathbb{S}_p} D_{uu}\widetilde{f}^\mu(p). \tag{21}$$

Thus we have

$$\begin{aligned}&\min_{u\in\mathbb{S}_p} D_{uu}\widetilde{f}^\mu(p) \\ =& \min_{u\in\mathbb{S}_p}\left\{\frac{1}{2\mu}\mathbb{E}_{v\sim\mathbb{S}_p}\left[\int_{-\mu}^{\mu} D_{u_{\mathrm{Exp}_p(tv)}u_{\mathrm{Exp}_p(tv)}} f\left(\mathrm{Exp}_p(tv)\right)dt + \frac{1}{2}\mu^3\left\langle \mathrm{grad} f\big|_p, \nabla_v R(v,u)u + \nabla_u R(v,u)v\right\rangle\right]\right\} + O\left(\mu^5\right) \\ \geq & \frac{1}{2\mu}\mathbb{E}_{v\sim\mathbb{S}_p}\left[\int_{\mu}^{\mu}\min_{w\in\mathbb{S}_{\mathrm{Exp}_p(tv)}} D_{ww} f\left(\mathrm{Exp}_p(tv)\right)dt\right] + \min_{u\in\mathbb{S}_p}\frac{1}{2}\mu^3\mathbb{E}_{v\sim\mathbb{S}_p}\left[\left\langle \mathrm{grad} f\big|_p, \nabla_v R(v,u)u + \nabla_u R(v,u)v\right\rangle\right] + O\left(\mu^5\right).\end{aligned}$$

Since $\lambda_{\min}\left(H_{p,\widetilde{f}^\mu}\right) = \min_{u\in\mathbb{S}_p} u^\top H_{p,\widetilde{f}^\mu} u = \min_{u\in\mathbb{S}_p} D_{uu}\widetilde{f}^\mu(p)$ for any p and f, it holds that

$$\begin{aligned}\lambda_{\min}\left(H_{p,\widetilde{f}^\mu}\right) \geq & \frac{1}{2\mu}\mathbb{E}_{v\sim\mathbb{S}_p}\left[\int_{-\mu}^{\mu}\lambda_{\min}\left(H_{\mathrm{Exp}_p(tv),f}\right)dt\right] \\ & + \min_{u\in\mathbb{S}_p}\frac{1}{2}\mu^3\mathbb{E}_{v\sim\mathbb{S}_p}\left[\left\langle \mathrm{grad} f\big|_p, \nabla_v R(v,u)u + \nabla_u R(v,u)v\right\rangle\right] + O\left(\mu^5\right).\end{aligned}$$

□

3 Estimating Gradient over Riemannian Manifolds

From our formulation of the GW approximation, one get obtain tighter bounds for gradient estimation over Riemannian manifolds, for geodesically L_1-smooth functions defined as follows.

Definition 3. *Let $p \in \mathcal{M}$. A function $f : \mathcal{M} \to \mathbb{R}$ is called geodesically L_1-smooth near p if*

$$\left\|\mathcal{P}_{p\to q}\left(\mathrm{grad} f\big|_p\right) - \mathrm{grad} f\big|_q\right\|_q \leq L_1 d_p(p,q), \quad \forall q \in U_p, \tag{22}$$

where $\mathrm{grad} f\big|_p$ is the gradient of f at p. If f is L_1-smooth near p for all $p \in \mathcal{M}$, we say f is L_1-smooth over $\mathcal{M}$.

For a geodesically L_1-smooth function, one has the following theorem that bridges the gradient at p and the zeroth-order information near p.

Theorem 4. *Fix $\alpha > 0$. Let f be geodesically L_1-smooth near p. It holds that*

$$\left\| \mathrm{grad} f\big|_p - \frac{n}{\mu\alpha^2} \mathbb{E}_{v\sim\mathbb{S}_p(\alpha)} \left[f\left(\mathrm{Exp}_p(\mu v)\right) v \right] \right\|_p \leq \frac{L_1 n \alpha \mu}{2}, \tag{23}$$

where $\mathbb{S}_p(\alpha)$ is the origin-centered sphere in $T_p\mathcal{M}$ of radius α.

This theorem provides a very simple and practical gradient estimation method: At point $p \in \mathcal{M}$, we can select small numbers μ and α, and uniformly sample a vector $v \sim \mathbb{S}_p(\alpha)$. With these μ, α, v, the random vector

$$\widehat{\mathrm{grad}} f\big|_p(v) = \frac{n}{2\alpha^2\mu} \left(f\left(\mathrm{Exp}_p(\mu v)\right) - f\left(\mathrm{Exp}_p(-\mu v)\right) \right) \tag{24}$$

gives an estimator of $\mathrm{grad} f\big|_p$.

One can also independently sample $v_1, v_2, v_3, \cdots, v_m$ uniformly from $\mathbb{S}_p$. Using these samples, an estimator for $\mathrm{grad} f\big|_p$ is

$$\widehat{\mathrm{grad}} f\big|_p(v_1, v_2, \cdots, v_m) = \frac{n}{2m\alpha^2\mu} \sum_{i=1}^{m} \left(f\left(\mathrm{Exp}_p(\mu v_i)\right) - f\left(\mathrm{Exp}_p(-\mu v_i)\right) \right) v_i. \tag{25}$$

Note that Theorem 4 also provide an in expectation bound for the estimator in (25). Before proving Theorem 4, we first present Propositions 2 and 3, and Lemma 1.

Proposition 2. *Consider a continuously differentiable function f defined over a complete Riemannian manifold $\mathcal{M}$. If function f is geodesically L_1-smooth over $\mathcal{M}$, then it holds that*

1. *$f\left(\mathrm{Exp}_p(u)\right) \leq f(p) + \left\langle \mathrm{grad} f\big|_p, u \right\rangle + \frac{L_1}{2}\|u\|^2$ for all $p \in \mathcal{M}$ and $u \in T_p\mathcal{M}$;*
2. *for any $v \in T_p\mathcal{M}$, $\left\| \mathcal{P}_{0,1}^{\gamma}\left(\mathrm{grad} f\big|_p\right) - \mathrm{grad} f\big|_{\mathrm{Exp}_p(v)} \right\| \leq L_1\|v\|$, where $\gamma(t) := \mathrm{Exp}_p(tv)$.*

Proof. For any $p \in \mathcal{M}$ and $u \in T_p\mathcal{M}$, let $q_0 = p, q_1, q_2, \cdots, q_N = \mathrm{Exp}_p(u)$ be a sequence of points such that $q_{i+1} \in U_{q_i}$ and $q_{i+1} = \mathrm{Exp}_{q_i}(\tau u_{q_i})$ for all $\tau > 0$ that is sufficiently small. Since the Levi-Civita connection preserves the Riemannian metric, we know $N\tau = 1$. At any q_i, we have

$$\begin{aligned}
&\left\langle \mathrm{grad} f\big|_{q_i}, u_{q_i} \right\rangle \\
=&\left\langle \mathrm{grad} f\big|_{q_i} - \mathcal{P}_{q_{i-1}\to q_i}\left(\mathrm{grad} f\big|_{q_{i-1}}\right), u_{q_i} \right\rangle + \left\langle \mathrm{grad} f\big|_{q_{i-1}}, u_{q_{i-1}} \right\rangle \\
\leq&\left\langle \mathrm{grad} f\big|_{q_{i-1}}, u_{q_{i-1}} \right\rangle + L_1\tau\|u\|^2 \\
\leq&\cdots \\
\leq&\left\langle \mathrm{grad} f\big|_p, u \right\rangle + iL_1\tau\|u\|^2,
\end{aligned} \tag{26}$$

where (26) uses the definition of geodesic L_1-smoothness.

Also, it holds that

$$\begin{aligned}
f(q_i) - f(q_{i-1}) =& \int_0^\tau \left\langle \mathrm{grad} f\big|_{\mathrm{Exp}_{q_{i-1}}(tu_{q_{i-1}})}, u_{\mathrm{Exp}_{q_{i-1}}(tu_{q_{i-1}})} \right\rangle dt && \text{(by Lemma 1)} \\
=& \int_0^\tau \left\langle \mathrm{grad} f\big|_{\mathrm{Exp}_{q_{i-1}}(tu_{q_{i-1}})} - \mathcal{P}_{q_{i-1}\to \mathrm{Exp}_{q_{i-1}}(tu_{q_{i-1}})}\left(\mathrm{grad} f\big|_{q_{i-1}}\right), u_{\mathrm{Exp}_{q_{i-1}}(tu_{q_{i-1}})} \right\rangle dt && (27) \\
&+ \tau \left\langle \mathrm{grad} f\big|_{q_{i-1}}, u_{q_{i-1}} \right\rangle \\
\leq& \tau \left\langle \mathrm{grad} f\big|_{q_{i-1}}, u_{q_{i-1}} \right\rangle + \frac{L_1\tau^2\|u\|^2}{2}.
\end{aligned}$$

where (27) uses the definition of geodesic L_1-smoothness.

Thus we have

$$
\begin{aligned}
f(\mathrm{Exp}_p(u)) - f(p) \leq & \sum_{i=1}^{N} f(q_i) - f(q_{i-1}) \\
\leq & \sum_{i=1}^{N} \left(\tau \left\langle \mathrm{grad} f\big|_{q_{i-1}}, u_{q_{i-1}} \right\rangle + \frac{L_1 \tau^2 \|u\|^2}{2} \right) \\
\leq & N\tau \left\langle \mathrm{grad} f\big|_p, u \right\rangle + \sum_{i=1}^{N} L_1 i \tau^2 \|u\|^2 + \frac{L_1 N \tau^2 \|u\|^2}{2} \\
= & \left\langle \mathrm{grad} f\big|_p, u \right\rangle + \frac{L_1 \|u\|^2}{2},
\end{aligned}
$$

where the last line uses that $\tau N = 1$. The above proves the first item.

The second item can be proved in a similar way. Specifically, we can also find a sequence of points $q_0, q_1, \cdots, q_N$ on the curve of $\gamma(t) = \mathrm{Exp}_p(tv)$ with $q_0 = p$ and $q_N = \mathrm{Exp}_p(v)$, and repeated use the definition of geodesic L_1-smoothness.

□

Proposition 3. *For any vector $u \in T_p\mathcal{M}$, we have*

$$
\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[\langle u, v \rangle v \right] = \frac{\alpha^2}{n} u, \quad \text{and} \quad \mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[\langle u, v \rangle^2 \right] = \frac{\alpha^2}{n} \|u\|^2.
$$

Proof. Since $T_p\mathcal{M}$ is a vector space, it is sufficient to consider $u = e_i$, where $\{e_i\}_i$ is the local canonical basis for $T_p\mathcal{M}$ with respect to g_p. In particular, we can choose $\{e_i\}_i$ such that $\langle e_i, e_j \rangle_p = \delta_{ij}$, where δ_{ij} is the Kronecker delta. For any $i, j \in \{1, 2, \cdots, n\}$,

$$
\left(\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[\langle e_i, v \rangle v \right] \right)_j = \mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[v_i v_j \right].
$$

Since $\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[v_j | v_i = x \right] = 0$ for any x when $i \neq j$, we have $\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[v_i v_j \right] = 0$. When $i = j$, by symmetry, we have $\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[v_j^2 \right] = \frac{\alpha^2}{n}$ for all $j = 1, 2, \cdots, n$.

We have shown $\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[\langle u, v \rangle v \right] = \frac{\alpha^2}{n} u$, on which taking inner product with u shows that $\mathbb{E}_{v \sim \mathbb{S}_p(\alpha)} \left[\langle u, v \rangle^2 \right] = \frac{\alpha^2}{n} \|u\|^2$. □

Lemma 1. *Pick $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$. Let $\gamma(t) = \mathrm{Exp}_p(tv)$ $(t \in [a, b])$ be a geodesic in $\mathcal{M}$. We then have*

$$
\int_a^b \left\langle \mathrm{grad} f\big|_{\mathrm{Exp}_p(sv)}, \dot{\gamma}\big|_s \right\rangle ds = f(\gamma(b)) - f(\gamma(a)). \tag{28}
$$

Proof. Consider the function $f \circ \gamma : [a, b] \to \mathbb{R}$. By fundamental theorem of calculus, we have

$$
f(\gamma(b)) - f(\gamma(a)) = \int_a^b \frac{df \circ \gamma}{dt} dt.
$$

Also, the directional derivative over Riemannian manifolds can be computed by

$$
\left. \frac{df \circ \gamma}{dt} \right|_s = \lim_{\tau \to 0} \frac{f \circ \gamma(s + \tau)}{\tau} = \left\langle \mathrm{grad} f\big|_{\mathrm{Exp}_p(sv)}, \dot{\gamma}\big|_s \right\rangle.
$$

Combining the above results finishes the proof. □

Lemma 1 can be viewed as a corollary of the fundamental theorem of calculus, or a consequence of Stokes' theorem. If one views the curve γ as a manifold with boundary ($\gamma(a)$ and $\gamma(b)$), and $\dot{\gamma}$ as a vector field parallel to γ, then applying Stokes' theorem proves Lemma 1.

We are now ready to prove Theorem 4, using our decomposition trick with the radial density function being $\Theta_{\mu,\alpha}(x) = \frac{1}{2\mu} \mathbb{I}_{[x \in [-\mu, \mu]]}$.

Proof of Theorem 4. From the definition of directional derivative, we have

$$\begin{aligned}\left\langle \mathrm{grad}\widetilde{f}_v^{\mu}\big|_p, v\right\rangle_p &= \frac{1}{2\mu}\lim_{\tau\to 0}\frac{\int_{-\mu}^{\mu}\Big(f\Big(\mathrm{Exp}_{\mathrm{Exp}_p(\tau v)}\Big(tv_{\mathrm{Exp}_p(\tau v)}\Big)\Big)-f\left(\mathrm{Exp}_p(tv)\right)\Big)dt}{\tau}\\ &= \frac{1}{2\mu}\int_{-\mu}^{\mu}\lim_{\tau\to 0}\frac{f\left(\mathrm{Exp}_p\left(\tau v+tv\right)\right)-f\left(\mathrm{Exp}_p\left(tv\right)\right)}{\tau}dt\\ &= \frac{1}{2\mu}\int_{-\mu}^{\mu}\left\langle \mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)}, v_{\mathrm{Exp}_p(tv)}\right\rangle_{\mathrm{Exp}_p(tv)}dt \qquad (29)\end{aligned}$$

For simplicity, let $\gamma_v(t)=\mathrm{Exp}_p(tv)$ ($t\in[-\mu,\mu]$) for any $v\in T_p(\mathcal{M})$. Since f is geodesically L_1-smooth, we have, for any p and $v\in T_p\mathcal{M}$,

$$\begin{aligned}&\left\langle \mathrm{grad}f\big|_p-\mathrm{grad}\widetilde{f}_v^{\mu}\big|_p, v\right\rangle_p\\ =&\left\langle \mathrm{grad}f\big|_p, v\right\rangle_p-\frac{\int_{-\mu}^{\mu}\left\langle \mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)}, v_{\mathrm{Exp}_p(tv)}\right\rangle_{\mathrm{Exp}_p(tv)}dt}{2\mu} \qquad (30)\\ =&\frac{1}{2\mu}\int_{-\mu}^{\mu}\left(\left\langle \mathcal{P}_{0,t}^{\gamma_v}\left(\mathrm{grad}f\big|_p\right), v_{\mathrm{Exp}_p(tv)}\right\rangle_{\mathrm{Exp}_p(tv)}\right)dt-\frac{1}{2\mu}\int_{-\mu}^{\mu}\left(\left\langle \mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)}, v_{\mathrm{Exp}_p(tv)}\right\rangle_{\mathrm{Exp}_p(tv)}\right)dt \qquad (31)\\ =&\frac{1}{2\mu}\int_{-\mu}^{\mu}\left(\left\langle \mathcal{P}_{0,t}^{\gamma_v}\left(\mathrm{grad}f\big|_p\right)-\mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)}, v_{\mathrm{Exp}_p(tv)}\right\rangle_{\mathrm{Exp}_p(tv)}\right)dt\\ \le&\frac{1}{2\mu}\int_{-\mu}^{\mu}L_1|t|\alpha^2dt=\frac{L_1\alpha^2\mu}{2}, \qquad (32)\end{aligned}$$

where (30) uses (29), (31) uses the fact that the parallel transport preserves the Riemannian metric, and (32) uses Proposition 2 and the Cauchy-Schwarz inequality.

By Proposition 3, we have

$$\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\left\langle \mathrm{grad}f\big|_p, v\right\rangle_p v\right]=\frac{\alpha^2}{n}\mathrm{grad}f\big|_p,$$

and thus from (32), we have

$$\begin{aligned}&\left\|\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\left\langle \mathrm{grad}\widetilde{f}_v^{\mu}\big|_p, v\right\rangle v\right]-\frac{\alpha^2}{n}\mathrm{grad}f\big|_p\right\|\\ =&\left\|\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\left\langle \mathrm{grad}\widetilde{f}_v^{\mu}\big|_p, v\right\rangle v-\left\langle \mathrm{grad}f\big|_p, v\right\rangle v\right]\right\|\\ =&\left\|\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\left\langle \mathrm{grad}\widetilde{f}_v^{\mu}\big|_p-\mathrm{grad}f\big|_p, v\right\rangle v\right]\right\|\\ \le&\frac{L_1\alpha^3\mu}{2}. \qquad (33)\end{aligned}$$

Applying Lemma 1 and the above results gives

$$\begin{aligned}&\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\left\langle \mathrm{grad}\widetilde{f}_v^{\mu}\big|_p, v\right\rangle_p v\right]\\ =&\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\frac{1}{2\mu}\int_{-\mu}^{\mu}\left\langle \mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)}, v_{\mathrm{Exp}_p(tv)}\right\rangle_{\mathrm{Exp}_p(tv)}dt\ v\right] && \text{(by Eq. 29)}\\ =&\frac{1}{2\mu}\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[\left(f\left(\mathrm{Exp}_p(\mu v)\right)-f\left(\mathrm{Exp}_p(\mu v)\right)\right)v\right] && \text{(by Lemma 1)}\\ =&\frac{1}{\mu}\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[f\left(\mathrm{Exp}_p(\mu v)\right)v\right].\end{aligned}$$

Combining the above result with (33) gives

$$\left\|\mathrm{grad}f\big|_p-\frac{n}{\mu\alpha^2}\mathbb{E}_{v\sim\mathbb{S}_p(\alpha)}\left[f\left(\mathrm{Exp}_p(\mu v)\right)v\right]\right\|\le\frac{L_1n\alpha\mu}{2}. \qquad (34)$$

□

The variance of the estimator (24) is shown in Theorem 5.

Theorem 5. *If f is geodesically L_1-smooth, then it holds that*

$$\begin{aligned}&\mathbb{E}_{v_1,v_2,\cdots,v_m\overset{i.i.d.}{\sim}\mathbb{S}_p(\alpha)}\left[\left\|\widehat{\mathrm{grad}}f\big|_p\left(v_1,v_2,\cdots,v_m\right)\big|_p\right\|^2\right]\\ \leq&\frac{n^2}{m}\left(\frac{L_1\alpha\mu}{2}+\left\|\mathrm{grad}f\big|_p\right\|\right)^2+\frac{m-1}{m}\left(\left\|\mathrm{grad}f\big|_p\right\|^2+\left\|\mathrm{grad}f\big|_p\right\|L_1n^{3/2}\alpha\mu+\frac{L_1^2n^3\alpha^2\mu^2}{4}\right).\end{aligned}$$

Proof. By Lemma 1 and that f is geodesically L_1-smooth, we have, for any $v\in\mathbb{S}_p(\alpha)$,

$$\begin{aligned}&\left|f\left(\mathrm{Exp}_p(\mu v)\right)-f\left(\mathrm{Exp}_p(-\mu v)\right)\right|\\ =&\left|\int_{-\mu}^{\mu}\left\langle\mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)},v_{\mathrm{Exp}_p(tv)}\right\rangle dt-\int_{-\mu}^{\mu}\left\langle\mathrm{grad}f\big|_p,v\right\rangle dt+2\mu\left\langle\mathrm{grad}f\big|_p,v\right\rangle\right|\\ \leq&\int_{-\mu}^{\mu}\left|\left\langle\mathrm{grad}f\big|_{\mathrm{Exp}_p(tv)}-\mathcal{P}_{p\to\mathrm{Exp}_p(tv)}\left(\mathrm{grad}f\big|_p\right),\mathcal{P}_{p\to\mathrm{Exp}_p(tv)}(v)\right\rangle\right|dt+\left|2\mu\left\langle\mathrm{grad}f\big|_p,v\right\rangle\right|\\ \leq&L_1\int_{-\mu}^{\mu}|t|\|v\|^2dt+2\alpha\mu\left\|\mathrm{grad}f\big|_p\right\|\\ \leq&L_1\alpha^2\mu^2+2\alpha\mu\left\|\mathrm{grad}f\big|_p\right\|.\end{aligned}\tag{35}$$

By (35), it holds that,

$$\begin{aligned}\mathbb{E}\left[\left\|\widehat{\mathrm{grad}}f\big|_p(v)\right\|^2\right]=&\mathbb{E}\left[\frac{n^2}{4\mu^2\alpha^4}\left(f\left(\mathrm{Exp}_p(\mu v)\right)-f\left(\mathrm{Exp}_p(-\mu v)\right)\right)^2\|v\|^2\right]\\ \leq&n^2\left(\frac{L_1\alpha\mu}{2}+\left\|\mathrm{grad}f\big|_p\right\|\right)^2\end{aligned}\tag{36}$$

By Theorem 4, we have

$$\mathbb{E}\left[\frac{n}{2\mu\alpha^2}\left(f\left(\mathrm{Exp}_p\left(\mu v_i\right)\right)-f\left(\mathrm{Exp}_p\left(-\mu v_i\right)\right)\right)v_i\right]=\mathrm{grad}f\big|_p+O\left(\frac{L_1n\alpha\mu}{2}\mathbf{1}\right),$$

where $\mathbf{1}$ is the all-one vector and the big-O notation here means that, with respect to any coordinate system, each entry in $\mathbb{E}\left[\frac{n}{2\mu\alpha^2}\left(f\left(\mathrm{Exp}_p\left(\mu v_i\right)\right)-f\left(\mathrm{Exp}_p\left(-\mu v_i\right)\right)\right)v_i\right]$ and $\mathrm{grad}f\big|_p$ differs by at most $\frac{L_1n\alpha\mu}{2}$.

Since $v_1,v_2,\cdots,v_m$ are mutually independent, we have, for any $i\neq j$,

$$\begin{aligned}&\mathbb{E}\left\langle\frac{n}{2\mu\alpha^2}\left(f\left(\mathrm{Exp}_p\left(\mu v_i\right)\right)-f\left(\mathrm{Exp}_p\left(-\mu v_i\right)\right)\right)v_i,\frac{n}{2\mu\alpha^2}\left(f\left(\mathrm{Exp}_p\left(\mu v_j\right)\right)-f\left(\mathrm{Exp}_p\left(-\mu v_j\right)\right)\right)v_j\right\rangle\\ =&\left\langle\mathrm{grad}f\big|_p+O\left(\frac{L_1n\alpha\mu}{2}\mathbf{1}\right),\mathrm{grad}f\big|_p+O\left(\frac{L_1n\alpha\mu}{2}\mathbf{1}\right)\right\rangle\\ \leq&\left\|\mathrm{grad}f\big|_p\right\|^2+\left\|\mathrm{grad}f\big|_p\right\|L_1n^{3/2}\alpha\mu+\frac{L_1^2n^3\alpha^2\mu^2}{4},\end{aligned}$$

where the last inequality uses the Cauchy-Schwartz inequality. Thus by expanding out all terms we have

$$\begin{aligned}&\mathbb{E}\left[\left\|\widehat{\mathrm{grad}}f\big|_p\left(v_1,v_2,\cdots,v_m\right)\right\|^2\right]\\ =&\frac{1}{m^2}\mathbb{E}\left[\left\|\sum_{i=1}^m\widehat{\mathrm{grad}}f\big|_p(v_i)\right\|^2\right]+\sum_{1\leq i,j\leq m:i\neq j}\mathbb{E}\left\langle\mathrm{grad}f\big|_p+O\left(\frac{L_1n\alpha\mu}{2}\mathbf{1}\right),\mathrm{grad}f\big|_p+O\left(\frac{L_1n\alpha\mu}{2}\mathbf{1}\right)\right\rangle\\ \leq&\frac{n^2}{m}\left(\frac{L_1\alpha\mu}{2}+\left\|\mathrm{grad}f\big|_p\right\|\right)^2+\frac{m-1}{m}\left(\left\|\mathrm{grad}f\big|_p\right\|^2+\left\|\mathrm{grad}f\big|_p\right\|L_1n^{3/2}\alpha\mu+\frac{L_1^2n^3\alpha^2\mu^2}{4}\right).\end{aligned}$$

□

3.1 Previous Methods for Riemannian Gradient Estimation

Previously, Nesterov and Spokoiny (2017); Li et al. (2020) have introduced gradient estimators using the following sampler. The estimator is

$$\widehat{\operatorname{grad}} f\big|_p(v) = \frac{f\left(\operatorname{Exp}_p(\mu v)\right) - f\left(\operatorname{Exp}_p(-\mu v)\right)}{2\mu} v, \tag{37}$$

where $v \sim \mathcal{N}(0, I)$ is a Gaussian vector, and $\mu > 0$ is a parameter controlling the width of the estimator. We use this Gaussian sampler as a baseline for empirical evaluations. For this estimator, Li et al. (2020) studied its properties over manifolds embedded in Euclidean space. Their result is in Proposition 4.

Proposition 4 (Li et al. (2020))**.** *Let $\mathcal{M}$ be an n-dimensional manifold embedded in a Euclidean space. If $f : \mathcal{M} \to \mathbb{R}$ is geodesically L_1-smooth over $\mathcal{M}$, then it holds that*

$$\left\| \mathbb{E}_{v\sim\mathcal{N}(0,I)} \left[\widehat{\operatorname{grad}} f\big|_p(v) \right] - \operatorname{grad} f\big|_p \right\| \leq \frac{L_1 (n+3)^{3/2} \mu}{2}$$

where $\widehat{\operatorname{grad}} f\big|_p(v)$ is the estimator in (37).

Remark 2. *If one picks $\alpha = \sqrt{n}$ in Theorem 4, one can obtain the error bound $\frac{L_1 n^{3/2}\mu}{2}$ for our method, and the error bound for the previous method is $\frac{L_1(n+3)^{3/2}\mu}{2}$ (Proposition 4). Why should we pick $\alpha = sqrtn$ in Theorem 4? This is because when $\alpha = \sqrt{n}$, we have $\mathbb{E}_{v\sim\mathbb{S}(\alpha)}[\|v\|^2] = \mathbb{E}_{v\sim\mathcal{N}(0,I)}[\|v\|^2] = n$. In words, when $\alpha = \sqrt{n}$, the random vectors used in our method (24) and the random vector used in the previous method (37) have the same squared norm in expectation, which leads to a fair comparison.*

4 Empirical Studies

In this section, we empirically study the spherical estimator in (24), in comparison with the Gaussian estimator in (37). The two methods (24) and (37) are compared over three manifolds: 1. the Euclidean space $\mathbb{R}^n$, 2. the unit sphere $\mathbb{S}^n$, and 3. the surface specified by the equation $h(x) = \frac{1}{2}\sum_{i=1}^{n/2} x_i^2 - \frac{1}{2}\sum_{i=n/2+1}^{n} x_i^2$. The evaluation is carried out at three different point, one for each manifold. The three manifolds, and corresponding points for evaluation, are listed in Table 1. Two test functions are used: a. the linear function $f(x) = \sum_{i=1}^n x_i$, and b. the function $f(x) = \sum_{i=1}^n \sin(x_i - 1)$. The two test functions are listed in Table 2.

Label	Manifold $\mathcal{M}$	Point p	Exponential map $\operatorname{Exp}_p(v)$
1	$\mathbb{R}^n$	0	v
2	$\mathbb{S}^n$	$(1, 0, 0, \cdots, 0)$	$p\cos(\|v\|) + \frac{v}{\|v\|}\sin(\|v\|)$
3	Surface of $h(x)$, $x \in \mathbb{R}^n$	0	$\left(v, \left(\frac{\sqrt{1+\sum_{i=1}^n v_i^4}}{2} + \frac{\sinh^{-1}\left(\sqrt{\sum_{i=1}^n v_i^4}\right)}{\sqrt{\sum_{i=1}^n v_i^4}}\right) h(v)\right)$

Table 1: Manifolds used for experiments. All manifolds are of dimension n. The coordinate system of the ambient space is used in all settings.

Label	Function
a	$f(x) = \sum_{i=1}^n x_i$
b	$f(x) = \sum_{i=1}^n \sin(x_i - 1)$

Table 2: Functions used for experiments. The coordinate system of the ambient Euclidean space is used for the computations.

The three manifolds in Table 1 and the two functions in Table 2 together generate 6 settings. We use 1a, 1b, 2a, 2b, 3a, 3c to label these 6 setting, where 1a refers to the setting over manifold 1 using function a, and so on. The experimental results are summarized in Figures 1-12. In all figures, "G" on the x-axis stands for the previous method using Gaussian estimator (37), and "S" on the x-axis stands for our method using the spherical estimator (24). Each violin plot summarizes $N = 100$ estimations, where each estimation uses $m = 100$ samples. To fairly compare (37) and (24), we use $\alpha = \sqrt{n}$ for all spherical estimators so that for

both the spherical estimator and the Gaussian estimator, one has $\mathbb{E}\left[\|v\|^2\right] = n$. The figure captions specify the settings used. For example, setting 1a is for function a in Table 2 over manifold 1 in Table 1. Below each subfigure, the values of n and μ are the dimension of the manifold and the parameter used in the estimators (both G and S).

Since $\mathbb{E}\left[\|v\|^2\right]$ are set to the same value for both G and S, we use $\left\|\widehat{\mathrm{grad}} f\big|_p(v_1, v_2, \cdots, v_m) - \mathrm{grad} f\big|_p\right\|$ to measure the error of the estimator, and $\left\|\widehat{\mathrm{grad}} f\big|_p(v_1, v_2, \cdots, v_m)\right\|^4$ to measure robustness of the estimator. For both the estimation error and the robustness, smaller values mean better performance. Figures 1-6 plot the robustness of the two estimators G and S, and Figures 7-12 show the corresponding errors. As shown in the figures, in all of the settings, the spherical estimator S is much more robust than the Gaussian estimator G, while achieving the same level of estimation error.

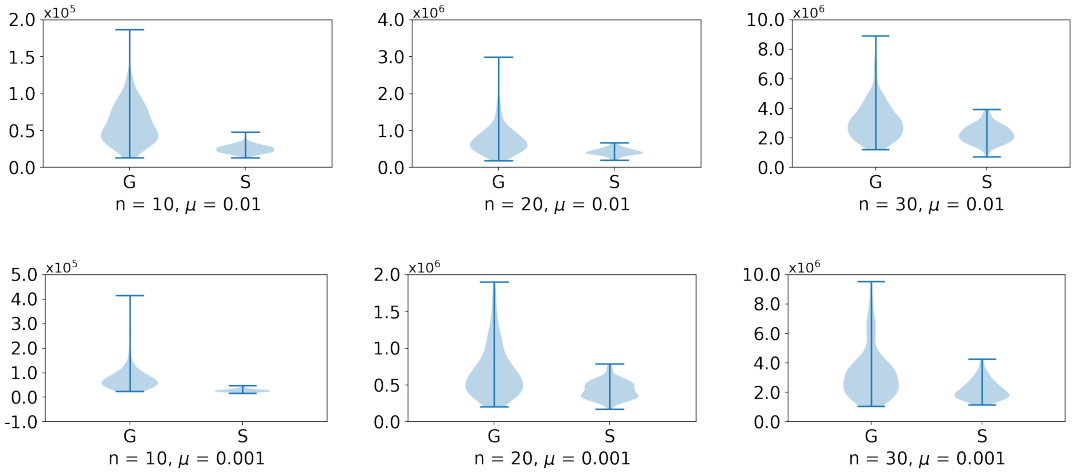

Figure 1: Plot of $\frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{\mathrm{grad}} f\big|_p(v_i)\right\|^4$ for Setting 1a.

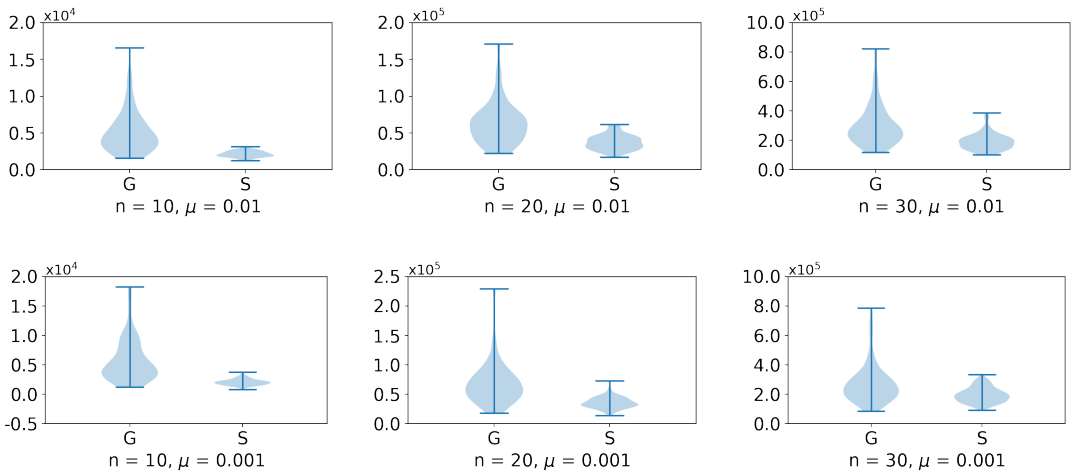

Figure 2: Plot of $\frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{\mathrm{grad}} f\big|_p(v_i)\right\|^4$ for Setting 1b.

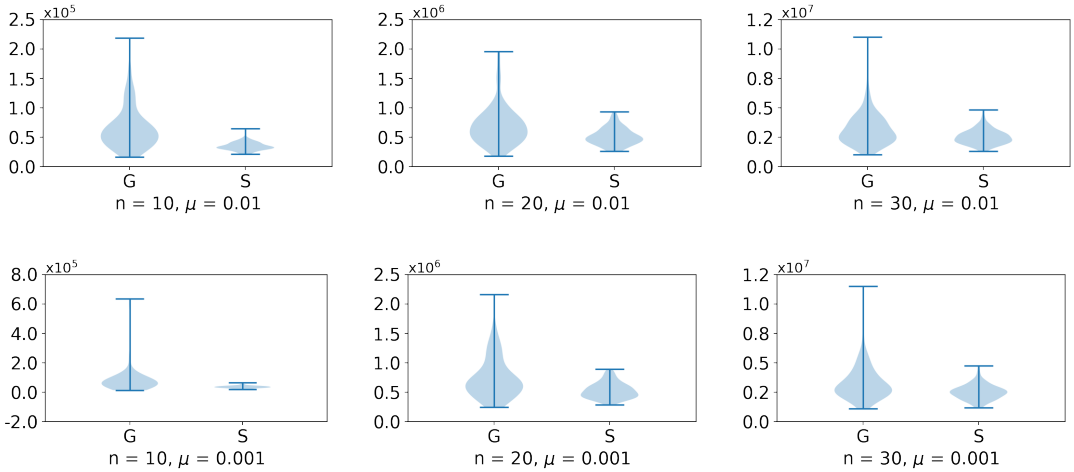

Figure 3: Plot of $\frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{\mathrm{grad}}f\big|_p(v_i)\right\|^4$ for Setting 2a.

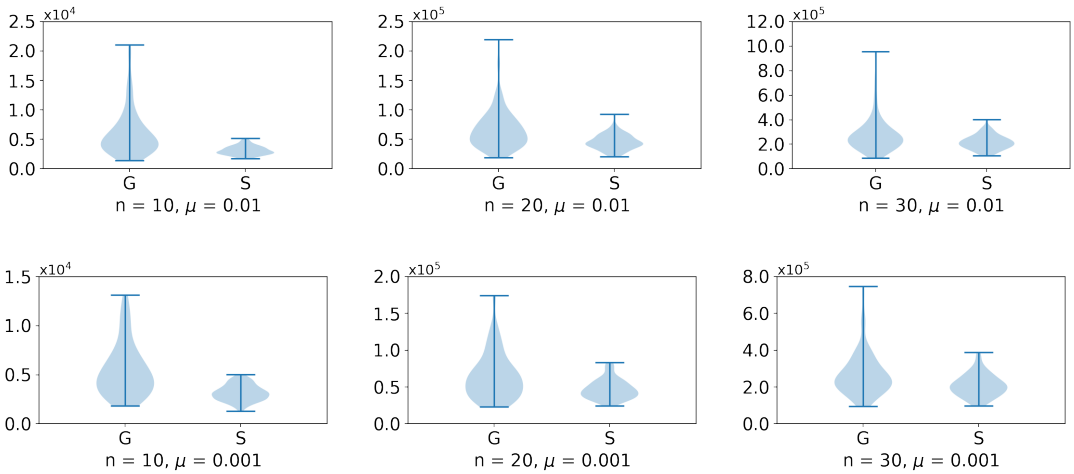

Figure 4: Plot of $\frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{\mathrm{grad}}f\big|_p(v_i)\right\|^4$ for Setting 2b.

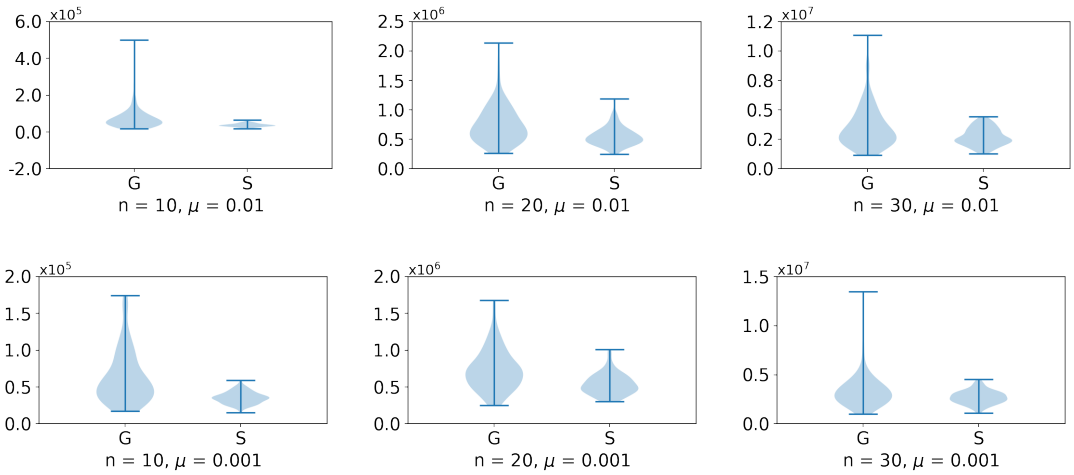

Figure 5: Plot of $\frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{\mathrm{grad}}f\big|_p(v_i)\right\|^4$ for Setting 3a.

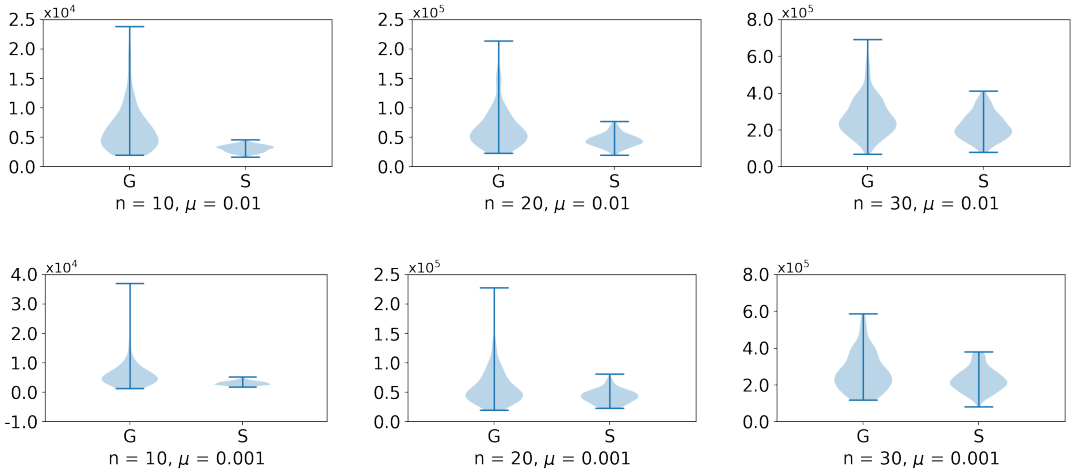

Figure 6: Plot of $\frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{\mathrm{grad}}f\big|_p(v_i)\right\|^4$ for Setting 3b.

Below are the plots of the estimation errors.

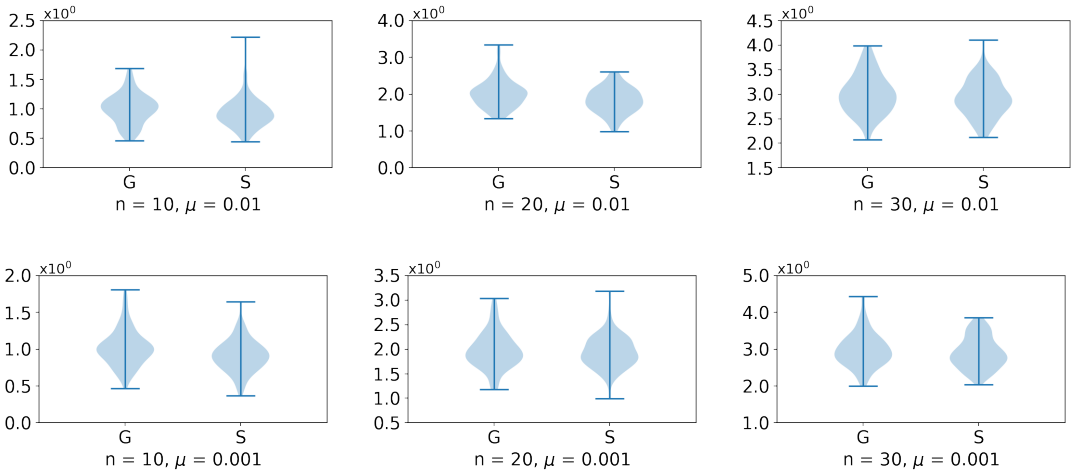

Figure 7: Plot of $\left\|\widehat{\mathrm{grad}}f\big|_p(v_1, v_2, \cdots, v_m) - \mathrm{grad}f\big|_p\right\|$ for Setting 1a.

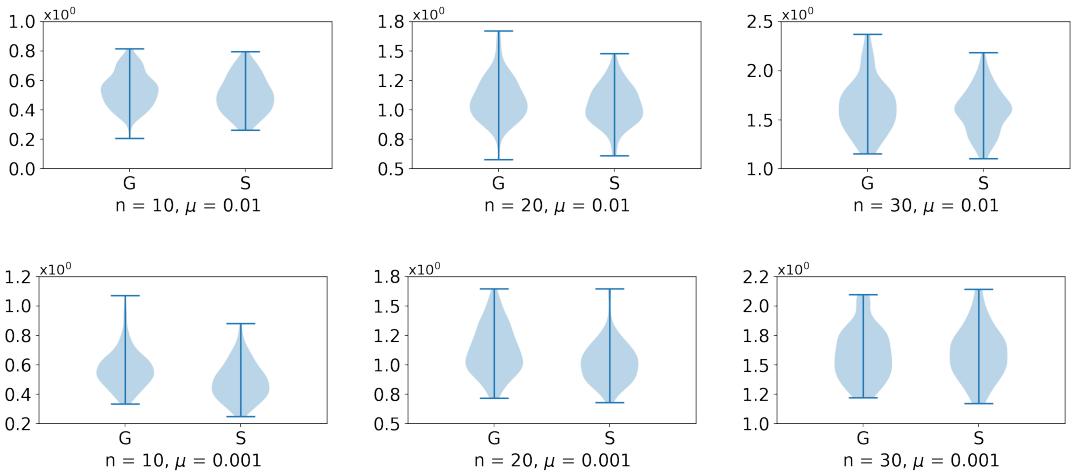

Figure 8: Plot of $\left\|\widehat{\mathrm{grad}}f\big|_p(v_1, v_2, \cdots, v_m) - \mathrm{grad}f\big|_p\right\|$ for Setting 1b.

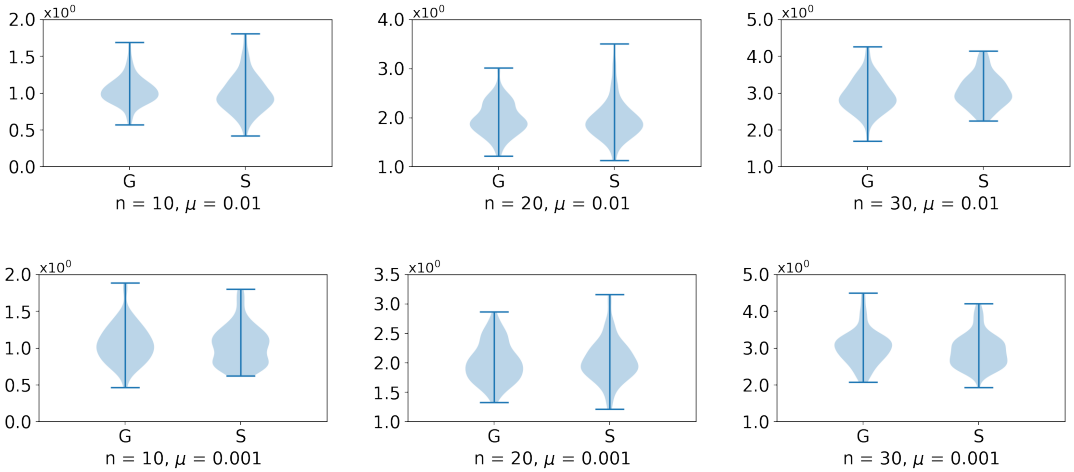

Figure 9: Plot of $\left\|\widehat{\mathrm{grad}} f\big|_p(v_1, v_2, \cdots, v_m) - \mathrm{grad} f\big|_p\right\|$ for Setting 2a.

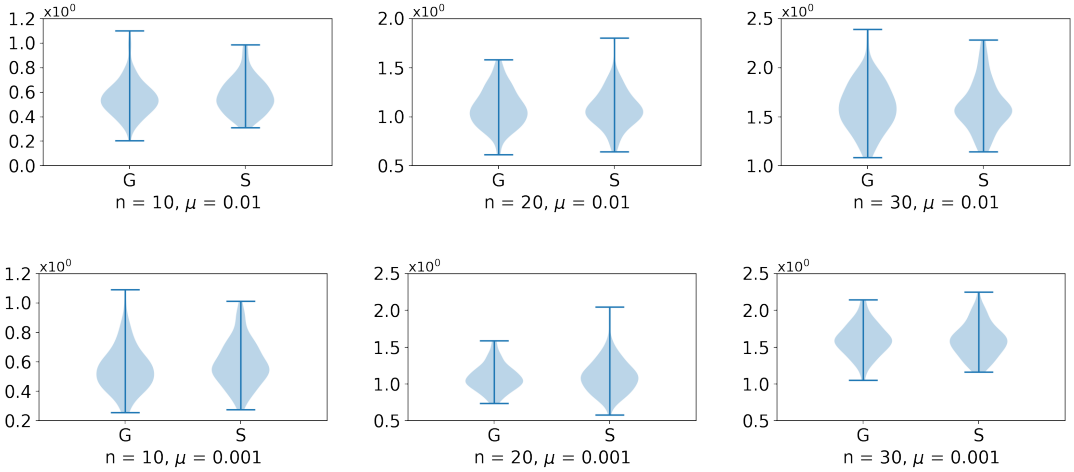

Figure 10: Plot of $\left\|\widehat{\mathrm{grad}} f\big|_p(v_1, v_2, \cdots, v_m) - \mathrm{grad} f\big|_p\right\|$ for Setting 2b.

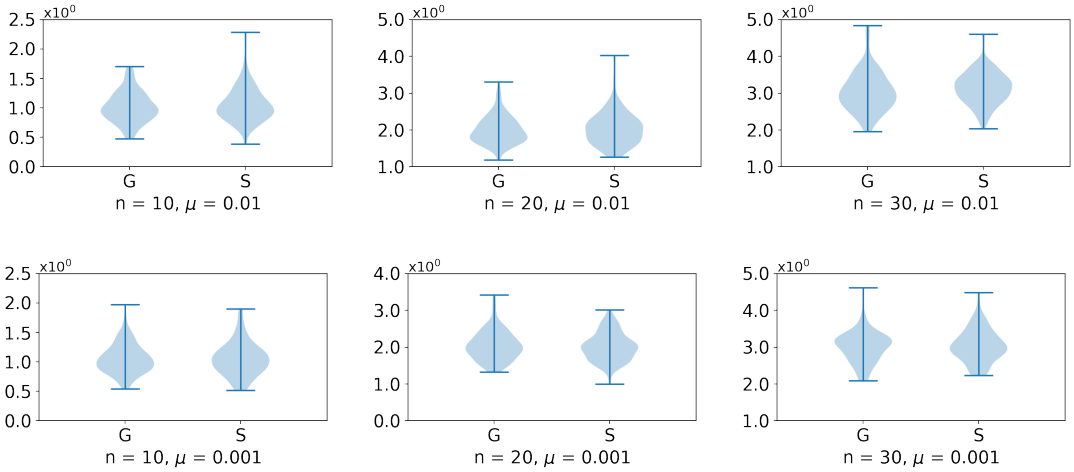

Figure 11: Plot of $\left\|\widehat{\mathrm{grad}} f\big|_p(v_1, v_2, \cdots, v_m) - \mathrm{grad} f\big|_p\right\|$ for Setting 3a.

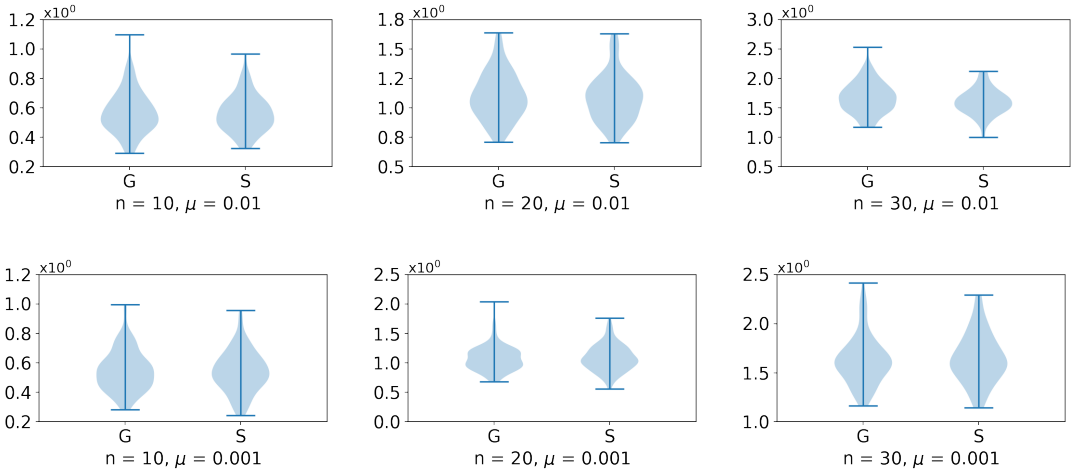

Figure 12: Plot of $\left\|\widehat{\text{grad}} f\big|_p(v_1, v_2, \cdots, v_m) - \text{grad} f\big|_p\right\|$ for Setting 3b.

5 Application to Bandit Convex Optimization

In online learning with bandit feedback, each time t, the learner chooses points x_t and z_t, and the environment picks a convex loss function f_t. The learner suffers loss $f_t(x_t)$ (not observed unless $z_t = x_t$) and observes the function value $f_t(z_t)$. The learner then picks x_{t+1} based on historical observations and the learning process moves on. In this learning protocol, when $z_t = x_t$, the learner observes and suffers loss both at x_t. Online learning agents, unlike their offline learning (or batch learning) counterpart, do not have knowledge about f_t when they choose x_t. Such models capture real-world scenarios such as learning with streaming data. In the machine learning community, such problem settings are often referred to as the *Bandit Convex Optimization* problem.

5.1 Previous Works on Bandit Convex Optimization

The problem of online convex optimization with bandit feedback was introduced and studied by Flaxman et al. (2005). Since then, a sequence of works has focused on bandit convex learning in Euclidean spaces. For example, Hazan and Kale (2012); Chen et al. (2019); Garber and Kretzu (2020) have studied the gradient-free counterpart for the Frank-Wolfe algorithm, and Hazan and Levy (2014) studied this problem with self-concordant barrier. An extension to this scheme is combining the methodology with a partitioning of the space (Kleinberg, 2004). By doing so, regret rate of $\mathcal{O}\left(\sqrt{T}\right)$ (here T is the time horizon) can be achieved in Euclidean spaces (Bubeck et al., 2015; Bubeck and Eldan, 2016). This rate is optimal in terms of T, as shown by Shamir (2013). While there has been extensive study of online convex optimization problems, they are all restricted to Euclidean spaces. In this paper, we generalize previous results from Euclidean space to Hadamard manifolds, using our new gradient estimation method.

5.2 Bandit Convex Optimization over Hadamard Manifolds: an Algorithm

We consider a bandit convex optimization problem over a negatively curved manifold. Let there be a Hadamard manifold $\mathcal{M}$ and a closed geodesic ball $\mathcal{D} \subseteq \mathcal{M}$. Each time t, the agent and the environment executes the above learning protocol, with f_t defined over $\mathcal{M}$ and $x_t \in \mathcal{D}$. The goal of the agent is to minimize the T-step hindsight regret: $Reg(T) := \sum_{t=1}^{T} f_t(x_t) - f_t(x^*)$, where $x^* \in \arg\min_{x\in\mathcal{D}} \sum_{t=1}^{T} f_t(x)$.

With the approximation guarantee in Theorem 4, one can solve this online learning problem. At each t, we can approximate gradient at x_t using the function value $f_t(z_t)$, and perform an online gradient descent step. In particular, we can generalize the bandit convex optimization algorithm (Flaxman et al., 2005) to Hadamard manifolds. By Theorem 4, we know

$$\text{grad} f_t\big|_{x_t} \approx \mathbb{E}_{v_t\sim\mathbb{S}_{x_t}}\left[\frac{n}{\mu} f_t\left(\text{Exp}_{x_t}\left(\mu v_t\right)\right) v_t\right].$$

With this gradient estimator, one can perform estimated gradient descent by

$$x_{t+\frac{1}{2}} \leftarrow \mathrm{Exp}_{x_t}\left(\frac{n}{\mu} f_t\left(\mathrm{Exp}_{x_t}\left(\mu v_t\right)\right) v_t\right).$$

To ensure that the algorithm stays in the geodesic ball $\mathcal{D}$, we need to project $x_{t+\frac{1}{2}}$ back to a subset of $\mathcal{D}$, which is

$$(1-\beta)\mathcal{D} := B(c(\mathcal{D}), (1-\beta)r(\mathcal{D})), \qquad \text{for some } \beta \in (0,1),$$

where $c(\mathcal{D})$ and $r(\mathcal{D})$ are the center and radius of the geodesic ball $\mathcal{D}$. Since the set $(1-\beta)\mathcal{D}$ is a geodesic ball in a Hadamard manifold, the projection onto it is well-defined. Thus we can project $x_{t+\frac{1}{2}}$ onto $(1-\beta)\mathcal{D}$ to get x_{t+1}, or let $x_{t+1} = \mathrm{Proj}_{(1-\beta)\mathcal{D}}\left(x_{t+\frac{1}{2}}\right)$. Now one iteration finishes and the learning process moves on to the next round. This learning algorithm is summarized in Algorithm 1.

Algorithm 1

Input: Algorithm parameter: μ, η, β; time horizon: T;
Initialization: Arbitrarily pick $x_1 \in \mathcal{D}$.
for $t = 1, 2, \ldots, T$ **do**
  Environment picks loss function f_t. The agent suffers loss $f_t(x_t)$.
  Sample $v_t \sim \mathbb{S}_{x_t}$.
  Observe function value $f_t(z_t)$, where $z_t = \mathrm{Exp}_{x_t}(\mu v_t)$, and compute $w_t = \frac{n}{\mu} f_t(z_t) v_t$. /* w_t is the gradient estimate. */
  Perform gradient update $x_{t+\frac{1}{2}} = \mathrm{Exp}_{x_t}(-\eta w_t)$.
  Project the point to set $(1-\beta)\mathcal{D}$, $x_{t+1} = \mathrm{Proj}_{(1-\beta)\mathcal{D}}\left(x_{t+\frac{1}{2}}\right)$.
    /* In practice, the projection can be approximated by a point in $(1-\beta)\mathcal{D}$ that is close to $x_{t+\frac{1}{2}}$. */
end for

5.3 Bandit Convex Optimization over Hadamard Manifolds: Analysis

Theorem 6. *If, for all $t \in [T]$, the loss function f_t satisfies that is geodesically μ-smooth, then the regret of Algorithm 1 satisfies:*

$$\mathbb{E}\left[Reg(T)\right] \leq \frac{D^2}{2\eta} + \frac{\zeta(\kappa, D)\eta n^2 G^2 T}{2\mu^2} + \left(\frac{D L_1 n \mu}{2} + 2\beta D\right) T, \tag{38}$$

where D is the diameter of $\mathcal{D}$, and $G := \max_{p\in\mathcal{D}} f(p)$.

In particular, when $\eta = \Theta\left(T^{-3/4}\right)$, $\mu = \Theta\left(T^{-1/4}\right)$ and $\beta = \Theta\left(T^{-1/4}\right)$, the regret satisfies $\mathbb{E}\left[Reg(T)\right] \leq \mathcal{O}\left(T^{3/4}\right)$. This sublinear regret rate implies that, when $f_1, f_2, \cdots, f_T$ are identical, the algorithm converges to the optimal point at a rate of order $\mathcal{O}\left(T^{-1/4}\right)$. We now start the proof of Theorem 6, by presenting an existing result by Zhang and Sra.

Lemma 2 (Corollary 8, Zhang and Sra (2016))**.** *Let $\mathcal{M}$ be a Hadamard manifold, with sectional curvature $\mathcal{K}_p(u,v) \geq \kappa > -\infty$ for all $p \in \mathcal{M}$ and $u, v \in T_p\mathcal{M}$. Then for any x, x_s, and $x_{s+1} = \mathrm{Exp}_{x_s}(-\eta v)$ for some $v \in T_{x_s}\mathcal{M}$, then it holds that*

$$\left\langle -v, \mathrm{Exp}_{x_s}^{-1}(x)\right\rangle_{x_s} \leq \frac{1}{2\eta}\left(d^2(x_s, x) - d^2(x_{s+1}, x)\right) + \frac{\zeta(\kappa, d(x_s, x))\eta}{2}\|v\|^2,$$

where $\zeta(\kappa, D) = \frac{\sqrt{|\kappa|}D}{\tanh(\sqrt{|\kappa|}D)}$.

Proof of Theorem 6. By convexity of f_t, we have $f_t(x_t) - f_t(x^*) \leq \left\langle -\mathrm{grad} f_t\big|_{x_t}, \mathrm{Exp}_{x_t}^{-1}(x^*)\right\rangle$. Together with

the Lemma of Zhang and Sra, we get

$$
\begin{aligned}
& f_t(x_t) - f_t(x^*) \\
\leq & \left\langle -\mathrm{grad} f_t\big|_{x_t}, \mathrm{Exp}_{x_t}^{-1}(x^*) \right\rangle \\
\leq & \left\langle -w_t, \mathrm{Exp}_{x_t}^{-1}(x^*) \right\rangle + \left\langle w_t - \mathrm{grad} f_t\big|_{x_t}, \mathrm{Exp}_{x_t}^{-1}(x^*) \right\rangle \\
\leq & \frac{1}{2\eta}\left(d^2(x_t, x^*) - d^2(x_{t+\frac{1}{2}}, x^*)\right) + \frac{\zeta(\kappa, D)\eta}{2}\|w_t\|^2 \\
& + \left\langle w_t - \mathrm{grad} f_t\big|_{x_t}, \mathrm{Exp}_{x_t}^{-1}(x^*) \right\rangle,
\end{aligned}
$$

where the last line uses Lemma 2. Taking expectation (over $v_t \sim \mathbb{S}_{x_t}$) on both sides gives

$$
\begin{aligned}
& \mathbb{E}\left[f_t(x_t)\right] - \mathbb{E}\left[f_t(x^*)\right] \\
\leq & \frac{1}{2\eta}\mathbb{E}\left[d^2(x_t, x^*) - d^2(x_{t+\frac{1}{2}}, x^*)\right] + \mathbb{E}\left[\frac{\zeta(\kappa, D)\eta}{2}\|w_t\|^2\right] \\
& + \left\langle \mathbb{E}\left[w_t\right] - \mathrm{grad} f_t\big|_{x_t}, \mathrm{Exp}_{x_t}^{-1}(x^*) \right\rangle \\
\leq & \frac{\mathbb{E}\left[d^2(x_t, x^*) - d^2(x_{t+\frac{1}{2}}, x^*)\right]}{2\eta} + \frac{\zeta(\kappa, D)\eta n^2 G^2}{2\mu^2} + \frac{D L_1 n \mu}{2},
\end{aligned} \tag{39}
$$

where the last line uses Theorem 4, and that $\|w_t\|^2 \leq \frac{n^2 G^2}{\mu^2 \alpha^2}$ ($G \geq f(p)$ for all $p \in \mathcal{D}$). Henceforth, we will omit the expectation operator $\mathbb{E}$ for simplicity.

By definition of $(1-\beta)\mathcal{D}$, we have, $d\left(x_{t+\frac{1}{2}}, x_{t+1}\right) \leq \beta$. Plugging this back to (39), we get, in expectation,

$$
\begin{aligned}
f_t(x_t) - f_t(x^*) \leq & \frac{1}{2\eta}\left(d^2(x_t, x^*) - d^2(x_{t+1}, x^*)\right) + \frac{\zeta(\kappa, D)\eta n^2 G^2}{2\mu^2} + \frac{D L_1 n \mu}{2} + d^2(x_{t+1}, x^*) - d^2(x_{t+\frac{1}{2}}, x^*) \\
\leq & \frac{1}{2\eta}\left(d^2(x_t, x^*) - d^2(x_{t+1}, x^*)\right) + \frac{\zeta(\kappa, D)\eta n^2 G^2}{2\mu^2} + \frac{D L_1 n \mu}{2} + 2\beta D.
\end{aligned}
$$

Summing over t gives, in expectation,

$$
\begin{aligned}
& \sum_{t=1}^{T} f_t(x_t) - f_t(x^*) \\
\leq & \frac{1}{2\eta}\sum_{t=1}^{T}\left(d^2(x_t, x^*) - d^2(x_{t+1}, x^*)\right) + \frac{\zeta(\kappa, D)\eta n^2 G^2 T}{2\mu^2} + \left(\frac{D L_1 n \mu}{2} + 2\beta D\right) T \\
\leq & \frac{D^2}{2\eta} + \frac{\zeta(\kappa, D)\eta n^2 G^2 T}{2\mu^2} + \left(\frac{D L_1 n \mu}{2} + 2\beta D\right) T,
\end{aligned}
$$

where the last line telescopes out the summation term.

□

6 Conclusion

In this paper, we study the GW convolution/approximation and gradient estimation over Riemannian manifolds. Our new decomposition trick of the GW approximation provides a tools for computing properties of the GW convolution. Using this decomposition trick, we derive a formula for how the curvature of the manifold will affect the curvature of the function via the GW convolution. Empowered by our decomposition trick, we introduce a new gradient estimation method over Riemannian manifolds. Over an n-dimensional Riemannian manifold, we improve best previous gradient estimation error (Li et al., 2020) from $\mathcal{O}\left((n+3)^{3/2}\right)$ to $\mathcal{O}\left(n^{3/2}\right)$. As evidenced by a thorough empirical evaluation, our method is supreme to the best existing method. We also apply our gradient estimation method to bandit convex optimization problems, and generalize previous results from Euclidean spaces to Hadamard manifolds.

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